

Maths TESTName: Sacini Apurva~~Class: X~~ Sec: GrRoll: Batch: B10N1RDClass: X

- 1) (1) 3
- 2) (4) No solution
- 3) (1) 10 unit
- 4) (2) proportional
- 5) (2) $-\frac{1}{2}$
- 6) (1) x^3y^3

7) $p(x) = 4x^2 - 2x + (k-4)$
 $a=4, b=-2, c=k-4$

~~$\alpha + \beta = \frac{1}{2}$~~ Let α and β be zeros of $p(x)$

$$\alpha + \beta = \frac{1}{2} \alpha \beta \quad (\text{given})$$

$$\frac{-b}{a} = \frac{1}{2} \left(\frac{c}{a} \right) \quad \left[\text{by using } \alpha + \beta = \frac{-b}{a}, \alpha \beta = \frac{c}{a} \right]$$

$$\text{---} \left(\frac{1}{2} \right) \frac{-b}{a} = \frac{c}{2a}$$

$$\frac{-(-2)}{4} = \frac{k-4}{2(4)}$$

$$\frac{2}{4} = \frac{k-4}{8}$$

$$\frac{2 \times 8}{4} = k-4$$

$$4 = k-4$$

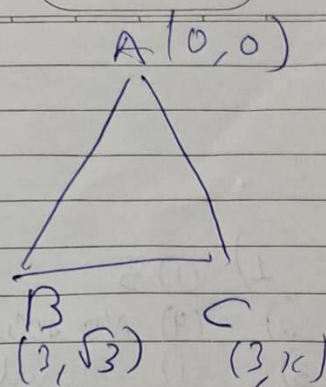
$$4+4 = k$$

$$\boxed{8 = k}$$

Ans: Value of k is 8

8) Since it's an equilateral triangle,

$$\begin{aligned}
 AB^2 &= AC^2 = BC^2 \\
 (3\sqrt{3})^2 &= 3^2 + x^2 + (\sqrt{3} - x)^2 \\
 27 &= 9 + x^2 + 3 + x^2 - 2\sqrt{3}x \\
 12 &= 2x^2 - 2\sqrt{3}x \\
 -2\sqrt{3}x &= -6 \\
 x &= \frac{3}{\sqrt{3}} = \sqrt{3}
 \end{aligned}$$



9)
$$\left. \begin{aligned} 2px + 3y &= 7 \\ 2x + y &= 6 \end{aligned} \right\} \text{ - given}$$

Given: To find: value of p

Solution: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (when there is unique solution)

By given equations

$$\begin{aligned}
 a_1 &= 2p, \quad b_1 = 3, \quad c_1 = 7 \\
 a_2 &= 2, \quad b_2 = 1, \quad c_2 = 6
 \end{aligned}$$

Substituting the values in $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

$$\frac{2p}{2} \neq \frac{3}{1}$$

$$\begin{aligned}
 2p &\neq 6 \\
 p &\neq 3
 \end{aligned}$$

So for all values of p except 3 it will have unique solution (one solution)

10) Given: $PD \parallel BC$,
 $\frac{AD}{DC} = \frac{CE}{BE}$

To Prove $PE \parallel AC$

Proof

Q. $PD \parallel BC$ (given)

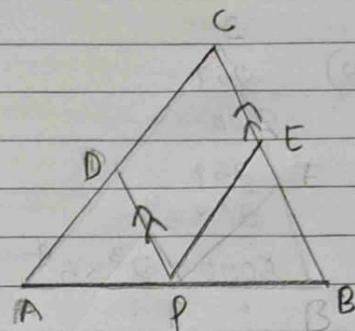
then $\frac{AD}{DC} = \frac{AP}{PB}$ (By BPT) $\rightarrow$ (1)

$\frac{AD}{DC} = \frac{CE}{BE}$ (given) $\rightarrow$ (2)

By (1) and (2)

$\frac{AP}{PB} = \frac{CE}{BE}$

$\therefore$ By converse of BPT
 $PE \parallel AC$



AD

AD

DC

RW

$\frac{\sin \theta}{\cos \theta} = \frac{1}{2}$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$

$\tan \theta = \frac{2 \sin \theta}{2 \cos \theta}$

$\frac{2 \sin \theta}{2 \cos \theta} = \frac{1}{2}$

$\frac{\sin \theta}{\cos \theta} = \frac{1}{2}$

$\sin^2 \theta = \cos^2 \theta$

Q

$\cos \theta = \sin \theta$

11) $\tan \theta = \frac{1}{2}$ $\leftarrow$ given

To Prove: $\frac{1}{2} \sin \theta \cos \theta = \frac{1}{5}$

Proof: $\tan \theta = \frac{1}{2}$ (given)

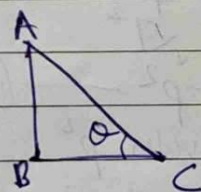
$\tan \theta = \frac{AB}{BC}$

$\frac{AB}{BC} = \frac{1}{2}$

Let K be nonzero common-multiple

$AB = K$ $BC = 2K$

Thus Proved



By Pythagoras Theorem

$AB^2 + BC^2 = AC^2$

$K^2 + (2K)^2 = AC^2$

$5K^2 = AC^2$

$\sqrt{5}K = AC$

LHS $= \frac{1}{2} \sin \theta \cos \theta$

$\frac{1}{2} \left(\frac{AB}{AC} \right) \left(\frac{BC}{AC} \right)$

$= \frac{1}{2} \left(\frac{1}{\sqrt{5}} \right) \left(\frac{2}{\sqrt{5}} \right) = \frac{1}{5}$

12)

$$\frac{257}{5000}$$

$$= \frac{257}{2^3 \times 5^4}$$

$$5000 = 2^3 \times 5^4$$

$$5000$$

$$\wedge$$

$$2 \times 2500$$

$$\wedge$$

$$5 \times 500$$

$$\wedge$$

$$5 \times 100$$

$$\wedge$$

$$5 \times 20$$

$$\wedge$$

$$5 \times 4$$

$$\wedge$$

$$2 \times 2$$

Decimal expansion is Terminating

13)

$$\sqrt{3} = \frac{p}{q}$$

Let $\sqrt{3}$ be a rational number

$\sqrt{3} = \frac{p}{q}$ (p and q are coprime integers where $q \neq 0$)

Squaring on both sides

$$(\sqrt{3})^2 = \left(\frac{p}{q}\right)^2$$

$$3 = \frac{p^2}{q^2}$$

$$3q^2 = p^2 \rightarrow (1)$$

3 divides p^2

3 is prime

$\therefore 3$ divides p

$\therefore 3$ is a factor of $p \rightarrow (2)$

$$p = 3k \rightarrow (3)$$

Substituting (3) in (1)

$$3q^2 = (3k)^2$$

$$3q^2 = 9k^2$$

$$q^2 = 3k^2$$

$\therefore 3$ divides q^2

$\therefore 3$ is prime

$\therefore 3$ divides q

$\therefore$ Therefore 3 is a factor of $q \rightarrow (4)$

By (2) and (4)

It can be said that p and q have a common factor 3

So this contradicts the fact that as these numbers are not coprime.

They are $\sqrt{3}$ is irrational number

14) $p(x) = 2x^2 - 3x + 7$

To find: $\frac{1}{2x-3} + \frac{1}{2\beta-3}$

Solution: $2x^2 - 3x + 7$

Comparing from $p(x)$

$a = 2, b = -3, c = 7$

$x + \beta = -\frac{b}{a}$

$x + \beta = -\frac{-3}{2}$

$x + \beta = \frac{3}{2} \rightarrow (1)$

$x\beta = \frac{c}{a}$

$x\beta = \frac{7}{2} \rightarrow (2)$

$\frac{1}{2x-3} + \frac{1}{2\beta-3}$

$\frac{2\beta-3 + 2x-3}{(2x-3)(2\beta-3)}$

$\frac{2\beta + 2x - 6}{2x\beta - 6x - 6\beta + 6}$

$\frac{2(x+\beta) - 6}{2x\beta - 6(x+\beta) + 6} \rightarrow (3)$

2k substituting values of ① and ② in ③

$$2k \left(\frac{3}{2} \right) - 6 = \frac{3-6}{7-9+6}$$

$$2 \left(\frac{7}{2} \right) - 6 \left(\frac{3}{2} \right) + 6$$

$$= \frac{-3}{-2+6}$$

$$= \frac{-3}{4}$$

$$\therefore \frac{1}{2x-3} + \frac{1}{2y-3} = \frac{-3}{4}$$

15) $3x - (a+1)y = 2b-1$
 $3x - (a+1)y - (2b-1) = 0 \longrightarrow \text{①}$

$$5x + (1-2a)y = 3b$$

$$5x + (1-2a)y - 3b = 0 \longrightarrow \text{②}$$

by ① and ②

$$a_1 = 3$$

$$b_1 = -(a+1)$$

$$c_1 = -(2b-1)$$

$$a_2 = 5$$

$$b_2 = 1-2a$$

$$c_2 = -3b$$

Given: They have infinitely many solutions.
 so,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$\frac{3}{5} = \frac{-(a+1)}{1-2a}$ Let's first take

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{3}{5} = \frac{-(a+1)}{1-2a}$$

$$\frac{3}{5} = \frac{-a-1}{1-2a}$$

$$3 - 6a = -5a - 5$$

$$3 + 5 = -5a + 6a$$

$$\boxed{8 = a}$$

Now let's take,

$$\frac{a_1}{a_2} = \frac{c_1}{c_2}$$

$$\frac{3}{5} = \frac{-(2b-1)}{+3b}$$

$$9b = 10b - 5$$

$$-b = -5$$

$$\boxed{b = 5}$$

$$\boxed{a = 8, b = 5}$$

16) Given: $\triangle ABE \cong \triangle ACD$
 To Prove: $\triangle ADE \sim \triangle ABC$
 Proof: $\triangle ABE \cong \triangle ACD$ (given)

$$AB = AC \text{ (C.P.C.T.)} \rightarrow (1)$$

$$AB = AD \text{ (C.P.C.T.)} \rightarrow (2)$$

By (1) and (2)

$$\frac{AB}{AC} = \frac{1}{1} \rightarrow (3)$$

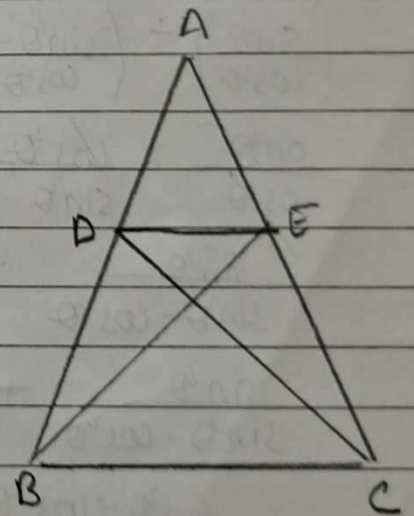
$$\frac{AD}{AE} = \frac{1}{1} \rightarrow (4)$$

By (3) and (4)

$$AB = AD$$

$$AC = AE$$

$$\frac{AB \times AB}{1} = \frac{AD \times AC}{1}$$



$$\frac{AB}{AD} = \frac{AC}{AE} \longrightarrow (5)$$

in $\triangle ADE$ and $\triangle ABC$

$$\frac{AB}{AD} = \frac{AC}{AE} \quad (\text{by } (5))$$

And $\angle DAE = \angle EAD$ (common)

$\triangle ADE \sim \triangle ABC$ (by SAS)

$$(7) \quad \frac{\tan^2 \theta}{\tan^2 \theta - 1} + \frac{\sec^2 \theta}{\sec^2 \theta - \cos^2 \theta} = \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

LHS =

$$\left(\frac{\sin^2 \theta}{\cos^2 \theta} \right) + \frac{1}{\sin^2 \theta}$$

$$\left(\frac{\sin^2 \theta}{\cos^2 \theta} - 1 \right) \left(\frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta} \right)$$

$$\left(\frac{\sin^2 \theta}{\cos^2 \theta} \right) \div \left(\frac{\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta} \right) + \frac{1}{\sin^2 \theta} \div \left(\frac{\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta \sin^2 \theta} \right)$$

$$\frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{1}{\sin^2 \theta} \times \frac{\cos^2 \theta \sin^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$\frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\cos^2 \theta \sin^2 \theta}{\sin^2 \theta (\sin^2 \theta - \cos^2 \theta)}$$

$$\frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$\frac{1}{\sin^2 \theta - \cos^2 \theta} = \text{LHS}$$

$$\text{RHS} = \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

Thus Proved

18) C. Ry section formula

$$C = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$(1,1) = \left[\frac{3x_2 + (2 \times -2)}{3+2}, \frac{3y_2 + (2 \times 7)}{3+2} \right]$$

$$(1,1) = \frac{3x_2 - 4}{5}, \frac{3y_2 + 14}{5}$$

$$1 = \frac{3x_2 - 4}{5}$$

$$1 = \frac{3y_2 + 14}{5}$$

$$5 = 3x_2 - 4$$

$$5 = 3y_2 + 14$$

$$3x_2 = 9$$

$$3y_2 = -9$$

$$x_2 = 3$$

$$y_2 = -3$$

19)

Let the numerator be x Let the denominator be y

According to first given condition,

$$\frac{2x}{y+1} = 1$$

$$2x = y + 1 \longrightarrow (1)$$

According to second given condition,

$$\frac{x+4}{2y} = \frac{1}{2}$$

$$2(x+4) = 2y$$

$$2x + 8 = 2y$$

$$x + 4 = y \longrightarrow (2)$$

$$x = y - 4 \longrightarrow (2)$$

Substituting (2) in (1)

$$\begin{aligned} 2(4-y) &= y+1 \\ 8-2y &= y+1 \\ -2y-y &= 1-8 \\ -3y &= -7 \\ y &= \frac{7}{3} \end{aligned}$$

$$\begin{aligned} 2(y-4) &= y+1 \\ 2y-8 &= y+1 \\ 2y-y &= 1+8 \\ y &= 9 \end{aligned} \longrightarrow (3)$$

Substituting (3) in (2)

$$x = y - 4$$

$$x = 9 - 4$$

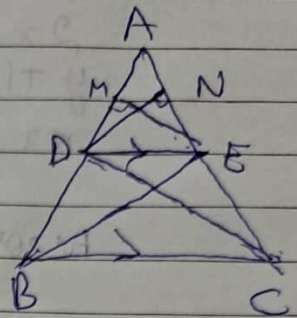
$$x = 5$$

Therefore fraction is $\frac{5}{9}$

20)

Basic Proportionality Theorem:

If a line is drawn parallel to one side of triangle intersecting at 2 distinct points of other 2 sides, then the line divides other 2 sides in the same ratio

Given $\triangle ABC$, $DE \parallel BC$ To prove: $\frac{AD}{DB} = \frac{AE}{EC}$ Construction : Draw $BM \perp AD$ Draw $DN \perp AE$ Join DC and EB Proof: In $\triangle ABC$

$$\text{ar}(\triangle) = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{ar}(\triangle ADE) = \frac{1}{2} \times AD \times EM \quad \text{--- (1)}$$

$$\text{ar}(\triangle DEB) = \frac{1}{2} \times DB \times EM \quad \text{--- (2)}$$

Dividing (1) by (2)

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DEB)} = \frac{\frac{1}{2} \times AD \times EM}{\frac{1}{2} \times DB \times EM} = \frac{AD}{DB} \quad \text{--- (3)}$$

$$\text{ar}(\triangle ADE) = \frac{1}{2} \times AE \times DN \longrightarrow (4)$$

$$\text{ar}(\triangle DEC) = \frac{1}{2} \times EC \times DN \longrightarrow (5)$$

Dividing (4) by (5)

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DEC)} = \frac{\frac{1}{2} \times AE \times DN}{\frac{1}{2} \times EC \times DN} = \frac{AE}{EC} \longrightarrow (6)$$

$$\text{ar}(\triangle DEB) = \text{ar}(\triangle DEC) \longrightarrow (7)$$

because they are on same base DE and between the same parallels.

by (3) (6) (7)

We can say that,

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Thus Proved

$$2) \frac{\sin 30 + 2\cos^2 45 + \tan^2 60}{\frac{1}{\cos 45} + \cos^2 30 + \tan^2 45}$$

$$= \frac{1 + 2\left(\frac{1}{\sqrt{2}}\right)^2 + (\sqrt{3})^2}{\frac{1}{2} + 1 + 3}$$

$$= \frac{\frac{1}{2} + \frac{3}{2} + 3}{\frac{1}{2} + \frac{3}{4} + 1}$$

$$= \frac{1 + 3 + 6}{2 + 3 + 4}$$

$$= \frac{10}{9}$$

$$= \left(\frac{1}{2} + 1 + 3 \right) \div \left(\frac{1}{2} + \frac{3}{4} + 1 \right)$$

$$= \left[\frac{1+2+6}{2} \right] \div \left[\frac{2+3+4}{4} \right]$$

$$= 2$$

~~22/02/2020~~

24)

Let the son's age be x years

Let the father's age be y years

According to first given condition,

$$2x + y = 70 \longrightarrow (1)$$

According to second given condition,

$$2y + 2x = 95 \longrightarrow (2)$$

Multiplying (1) by 2

$$2(2x + y) = 2(70)$$

$$4x + 2y = 140 \longrightarrow (3)$$

Subtracting (2) from (3)

$$4x + 2y = 140$$

$$x + 2y = 95$$

$$\begin{array}{r} (-) (-x) \quad (-) \end{array}$$

$$3x = 45$$

$$x = 15 \longrightarrow (4)$$

Age of son = 15

$$\left[\cos^3 \theta \right]^{2/3} + \left[\sin^3 \theta \right]^{2/3} \\ (\cos \theta)^{3 \times 2/3} + (\sin \theta)^{3 \times 2/3} \\ (\cos^2 \theta) + \sin^2 \theta = 1$$

$$RHS = 1$$

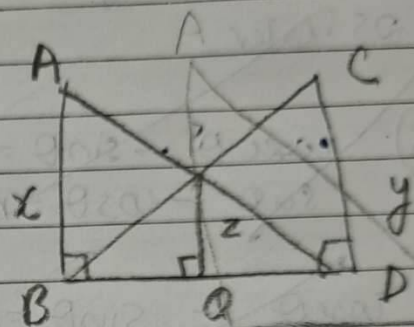
$$LHS = RHS$$

Thus Proved

30)

QED Given: $\angle ABD = \angle CPB = \angle PQB = 90^\circ$

To Prove: $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$



Proof: In $\triangle PQD$ and $\triangle ABD$

$$\angle PBD = \angle PQD \text{ (each } 90^\circ)$$

$$\angle D = \angle D \text{ (common)}$$

$$\triangle PQD \sim \triangle ABD \text{ (By AA)}$$

$$\frac{PQ}{AB} = \frac{QD}{BD} \text{ (By CEST)}$$

$$\frac{z}{x} = \frac{QD}{BD} \quad \text{--- (1)}$$

In $\triangle PQB$ and $\triangle CPB$

$$\angle CPB = \angle PQB \text{ (each } 90^\circ)$$

$$\angle B = \angle B \text{ (common)}$$

$$\triangle PQB \sim \triangle CPB \text{ (By AA)}$$

$$\frac{PQ}{CB} = \frac{QB}{PB} \text{ (By CEST)}$$

$$\frac{z}{y} = \frac{QB}{PB} \quad \text{--- (2)}$$

Adding (1) and (2)

$$26) \begin{aligned} \cos \theta - \sin \theta &= m \\ \sin \theta - \cos \theta &= n \end{aligned}$$

$$\text{To prove } (m^2 n)^{2/3} + (mn^2)^{2/3} = 1$$

$$\text{LHS} = [\cos \theta - \sin \theta]^2 [\sin \theta - \cos \theta]^{2/3} + [(\cos \theta - \sin \theta)(\sin \theta - \cos \theta)]^{2/3}$$

$$\left[\left(\frac{1}{\sin \theta} - \sin \theta \right)^2 \left[\frac{1}{\cos \theta} - \frac{\cos \theta}{\cos \theta} \right] \right]^{2/3} + \left[\left(\frac{1}{\sin \theta} - \sin \theta \right) \left(\frac{1}{\cos \theta} - \cos \theta \right) \right]^{2/3}$$

$$\left[\left(\frac{1 - \sin^2 \theta}{\sin \theta} \right)^2 \left(\frac{1 - \cos^2 \theta}{\cos \theta} \right)^{2/3} + \left[\left(\frac{1 - \sin^2 \theta}{\sin \theta} \right) \left(\frac{1 - \cos^2 \theta}{\cos \theta} \right)^2 \right]^{2/3} \right]$$

$$\left[\left(\frac{\cos^2 \theta}{\sin \theta} \right)^2 \cdot \left(\frac{\sin^2 \theta}{\cos \theta} \right)^{2/3} + \left[\left(\frac{\cos^2 \theta}{\sin \theta} \right) \left(\frac{\sin^2 \theta}{\cos \theta} \right)^2 \right]^{2/3} \right]$$

$$\left[\left(\frac{\cos^4 \theta \times \sin^2 \theta}{\sin^4 \theta \cos \theta} \right)^{2/3} + \left[\left(\frac{\cos^2 \theta}{\sin \theta} \right) \left(\frac{\sin^4 \theta}{\cos^2 \theta} \right) \right]^{2/3} \right]$$

$$\frac{z+z}{x \quad y} = \frac{QD + QB}{BD \quad BD}$$

$$\frac{z+z}{x \quad y} = \frac{BD}{BD}$$

$$\frac{z+z}{x \quad y} = 1$$

$$z \left(\frac{1}{x} + \frac{1}{y} \right) = 1$$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

Thus proved

29)

Let speed ~~be~~ of car A be x
Let speed of car B be y

According to given condition,

$$5x - 5y = 100 \longrightarrow (1)$$

According to second given condition,

$$x + y = 100 \longrightarrow (2)$$

Multiplying 5 ^{on} ~~in~~ ^{both} sides of equation (2)

$$5 \cdot x + 5y = 500 \longrightarrow (3)$$

~~Adding~~ Adding (1) and (3)

$$\begin{array}{r} 5x - 5y = 100 \\ 5x + 5y = 500 \\ \hline 10x = 600 \\ x = 60 \end{array}$$

Substituting $y = 60$
we get speed as 40